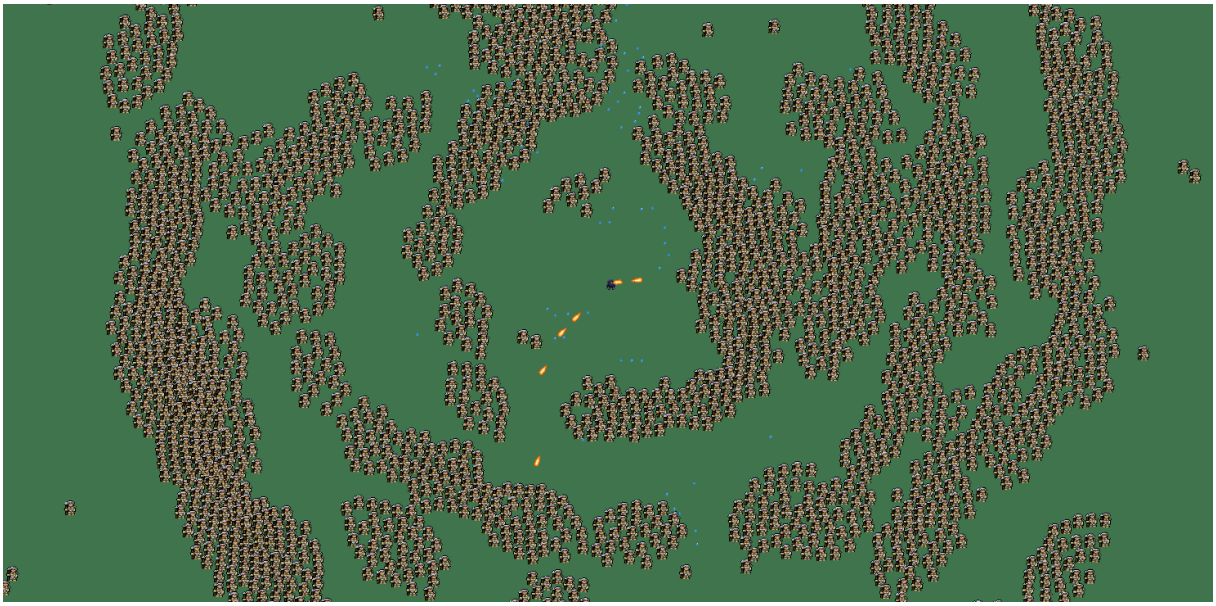
Roguelike: Performance Maxxing



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1. Context & Objectives

1.1 Context

As a lifelong video game enthusiast, I have recently transitioned from player to creator during my free time. This project is the combination of my passion for gaming and my academic background as a Software Engineering student.
The core motivation behind this development is to apply rigorous software engineering principles, such as modular architecture, clean code, and system optimization, to the complex environment of game development. By building a functional Roguelike from scratch, I aim to demonstrate how engineering patterns can solve the unique performance challenges inherent in "Survivor-like" games, where thousands of entities must be managed simultaneously.

Note that the main purpose of this project is not to make the most entertaining and fun game at first, I want to make a powerful game that could be played on almost any computer. I intend to implement fun features and make the game enjoyable after ending this project, follow my LinkedIn to see any potential announcement about the advancement of the game.

1.2 Objectives

The objective of this project is to fully compare the performances of a roguelike game using two fundamentally different architectures: Object-Oriented Programming (OOP) and DOTS/ECS.

- OOP Phase: The purpose is to find the limit where the player can have the most optimized and stable experience using standard Unity tools.
- DOTS/ECS Phase: We will then measure the performance gains with the new implementation and see how far we can push the limits of the game.

2. OOP

2.1 OOP Implementation Choices

Spatial grid: Reduced complexity from $O(n^2)$ to $O(1)$. To check collision of enemies with other enemies, you can just look at the cells around it.

Chunks: To make sure you can properly dezoom and move freely on the map, I implemented a chunk system. The chunks are loaded at the beginning and move with your character. As the game dezooms, it adds more chunks if needed to prevent the player from seeing outside the map.

Pools: Pools are used on entities that appear a lot on the screen, such as projectiles, enemies and experience. The pools allow you to allocate the necessary memory when the game starts and greatly reduces creation/destruction costs for each frame.

Observer: Observer Pattern on player to notify UI to update when necessary.

WeaponManager/ Scriptable Objects: WeaponManager handles the whole weapon logic, I added an interface IWeapon that every weapon inherits from, WeaponManager only has to call IWeapon.Attack() for each weapon in the player's inventory. The weapons are also ScriptableObjects and not MonoBehaviour, this allows you to add new weapons very efficiently with only a few modifications only in the Unity Interface.

Handle massive enemy horde movements (Boids Algorithm): To manage thousands of enemies on screen without severe jittering or overlapping, I implemented a custom Steering Behavior inspired by

the Boids algorithm. Instead of strictly applying instantaneous repulsive forces that cause visual "teleportation" and physics glitches, the enemies compute a blended directional vector. This blends a "Seek" behavior (moving toward the player) with a weighted "Separation" behavior (avoiding nearby neighbors). The separation vector is smoothly interpolated over time (Vector3.Lerp) to ensure a fluid, liquid-like horde movement. Additionally, the heavy separation calculations are time-sliced and distributed across multiple frames to significantly reduce CPU load.

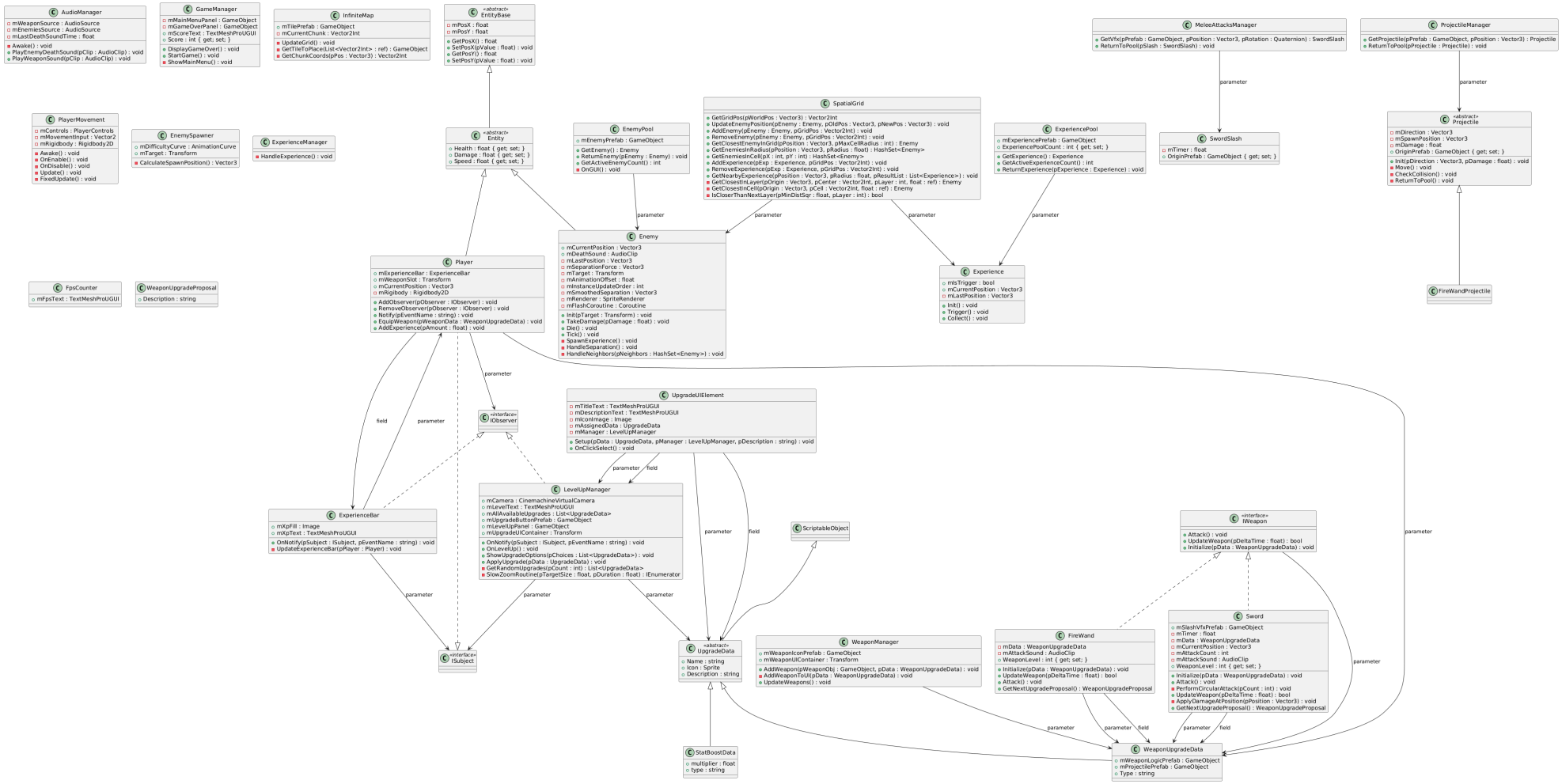
Clean Architecture:
Throughout the OOP phase, I made several adjustments to keep the architecture simple and the workflow efficient:

- **Removing Redundant Classes:** I realized that a custom EntityBase class to handle movements was unnecessary since Unity's built-in Transform component already handles positioning perfectly. Removing it simplified the entity hierarchy. I still kept it for the main attributes such as Health, Speed, Damage, etc…

- **Inspector Clean-up:** I systematically added [SerializeField] attributes to properly link UI elements directly within the Unity Inspector. This cleared up console warnings and made tweaking variables much easier.

- **Dynamic Item Drops:** I evolved the game design regarding loot. Instead of restricting items to spawn on fixed grid tiles, I modified the logic so they simply drop on the ground as physical objects, making the gameplay feel more dynamic.

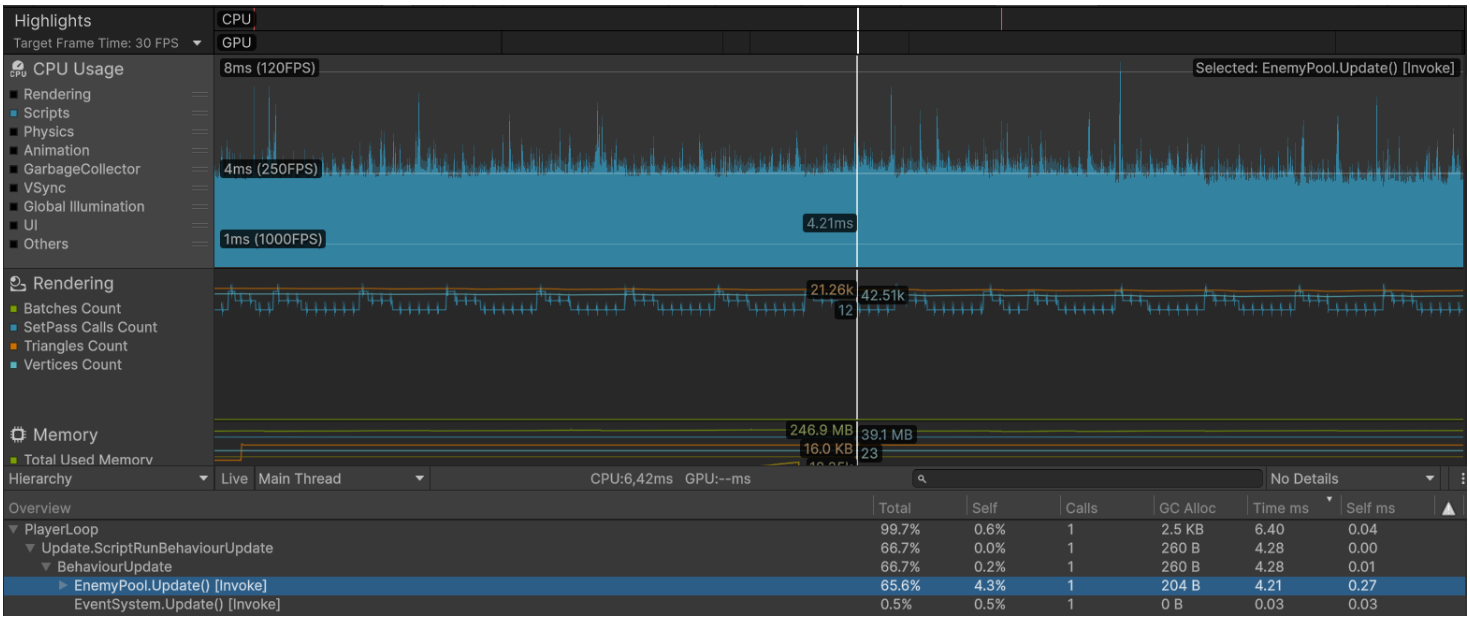
2.2 Class diagram



2.3 Performance analysis

Let's now try to find the limits of the OOP implementation. To begin with, I will analyze the code and try to find out what reduces the performance the most.

After using the profiler we see that the EnemyPool.Update() is what makes us lose the most performance:



We know the Update in EnemyPool calls the Tick () method in Enemy.cs:

```
void Update()
{
    for (int i = 0; i < mActiveEnemiesList.Count; i++)
    {
        mActiveEnemiesList[i].Tick();
    }
}
```

Now we know where to look at, let's go in the Tick () method and make a few changes at different parts so we can have more information in the profiler:

```
Profiler.BeginSample("1_Enemy_Maths_Boids");

mCurrentPosition = transform.position;
var lTargetPosition = mTarget.position;

// Compute the base direction towards the player
Vector3 lTargetDirection = (lTargetPosition - mCurrentPosition).normalized;

if ((Time.frameCount + mInstanceUpdateOrder) % 4 == 0) // Spread separation calculations over 4 frames to reduce CPU load
{
    HandleSeparation();
}

// Interpolate the separation force to prevent jittering and teleporting effects
mSmoothedSeparation = Vector3.Lerp(mSmoothedSeparation, mSeparationForce, Time.deltaTime * 10f);

// Combine target seeking and separation.
// Separation is weighted heavily (x4.0) to prioritize avoiding neighbors and prevent clumping
Vector3 lFinalDirection = (lTargetDirection + (mSmoothedSeparation * 4.0f)).normalized;

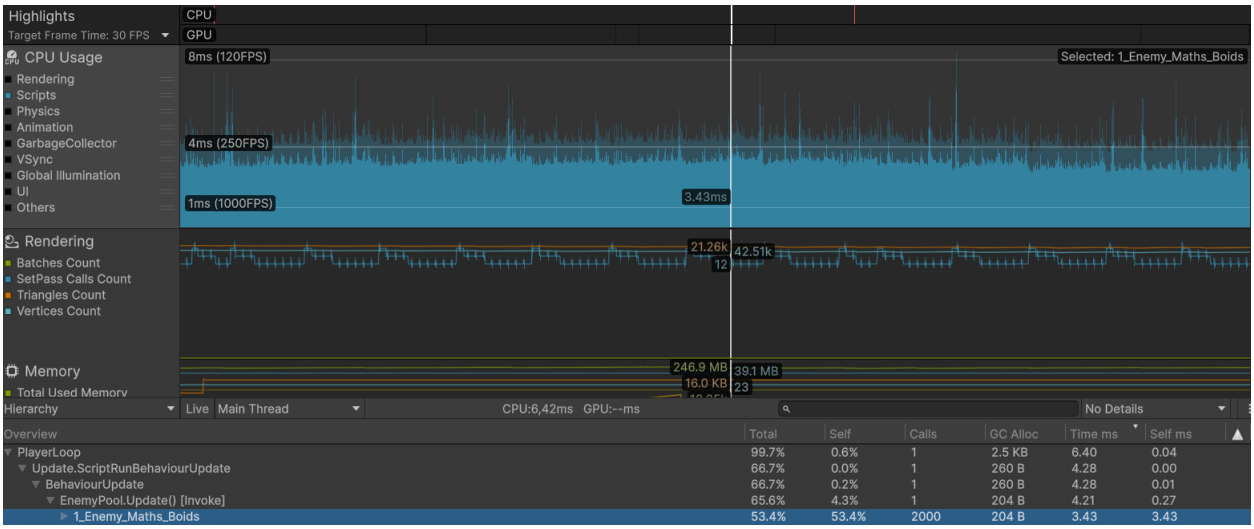
// Apply movement using the strict base speed to maintain a steady, fluid horde pace
mCurrentPosition += Speed * Time.deltaTime * lFinalDirection;

// Update the enemy's position within the spatial grid for accurate neighbor detection
SpatialGrid.Instance.UpdateEnemyPosition(this, mLastPosition, mCurrentPosition);
mLastPosition = mCurrentPosition;

// Apply a procedural tilt effect based on time to animate the sprite
float lTilt = Mathf.Sin((Time.time + mAnimationOffset) * mTiltSpeed) * mTiltAngle;

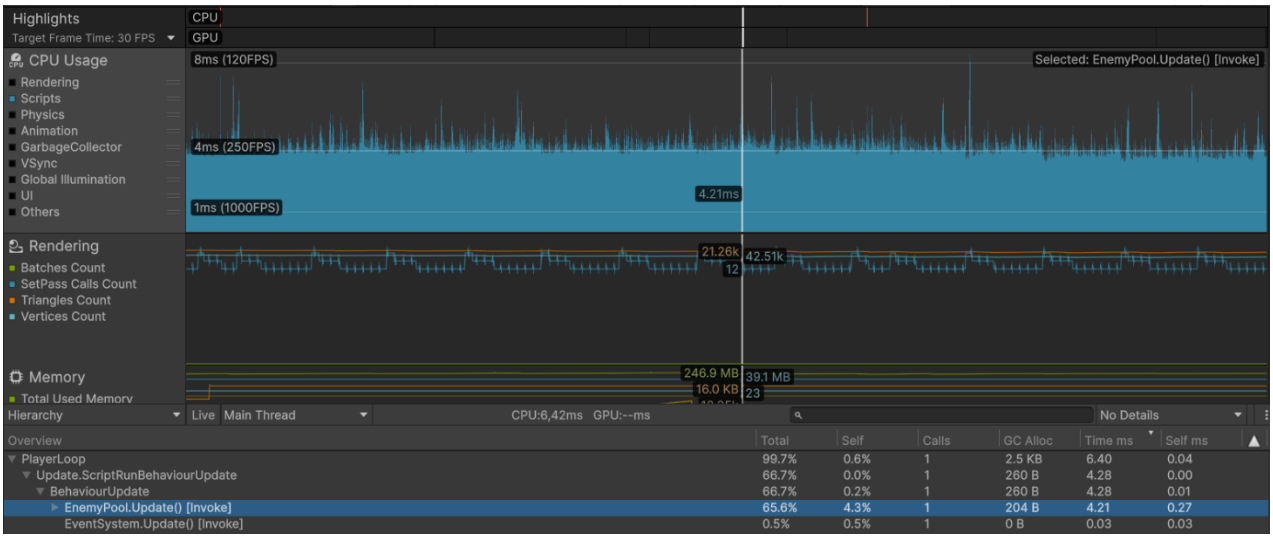
Profiler.EndSample();
```

Now let's go back to the profiler. We can now identify where the issue comes from. To no surprise, it comes from the Boids Algorithm, this is the part of the code that makes the most computations.



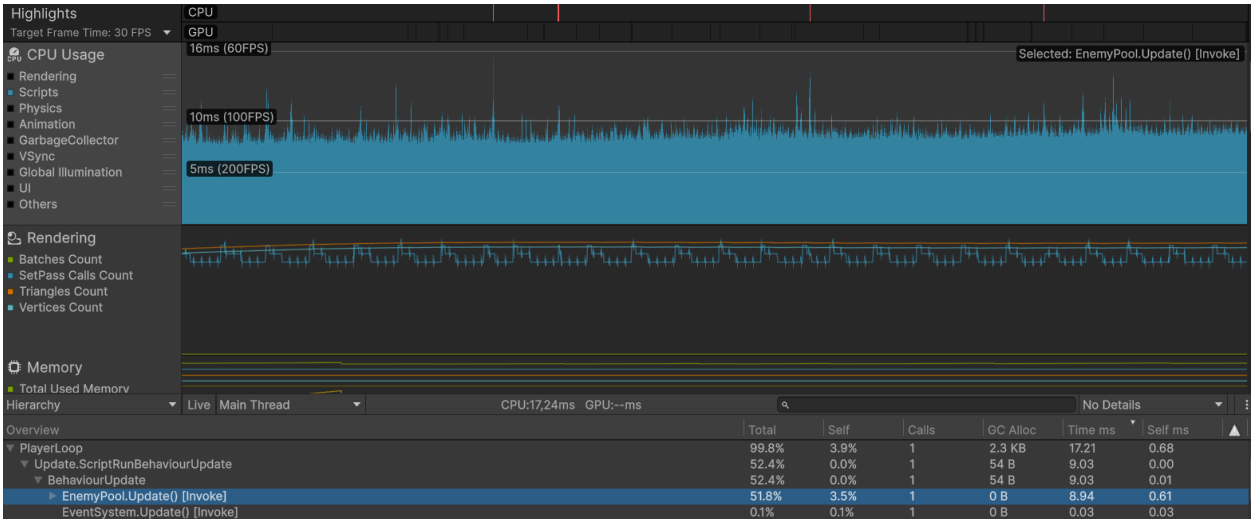
Let's now do a series of tests to see where the limit of this implementation is.

2.3.1 | 2000 Enemies



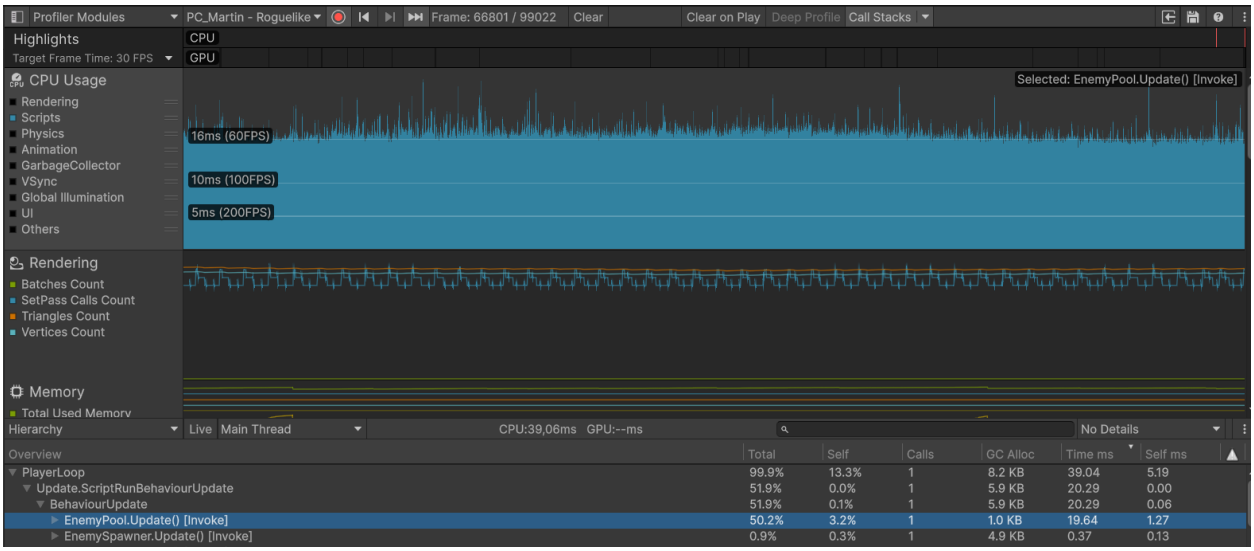
With 2000 enemies, we can see the update only takes 4.21ms and the in-game FPS is very stable above 144 fps.

2.3.2 | 4000 Enemies



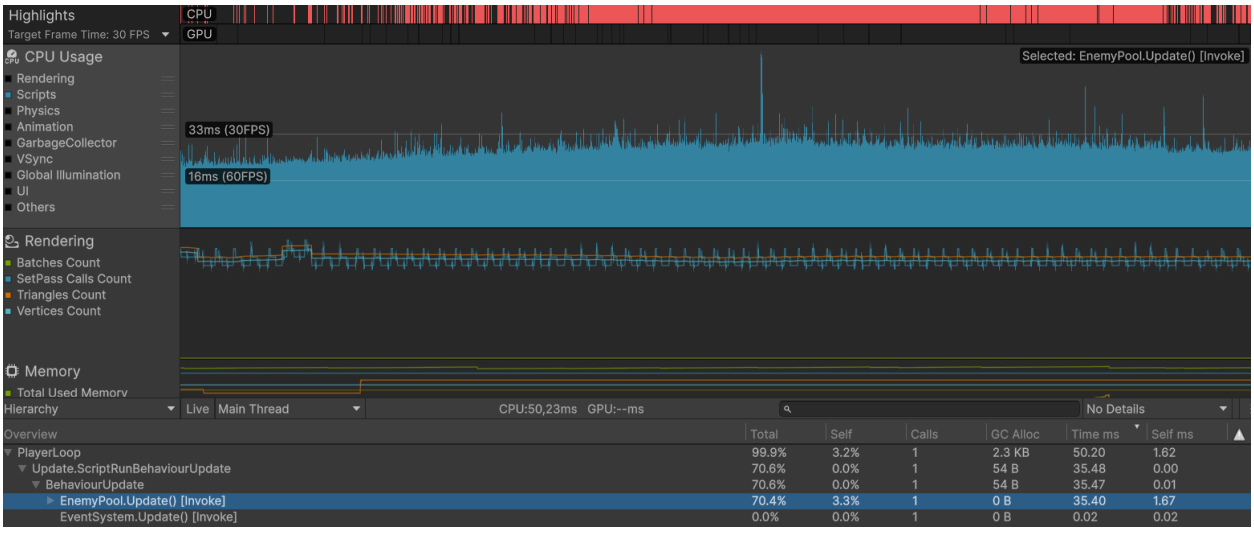
With 4000 enemies, we can already notice an important loss of performance, the method's time to compute more than doubled, going from 4.21ms to 8.94ms. In-game, the FPS starts dropping slightly below 144 fps, around 120 fps.

2.3.3 | 6000 Enemies



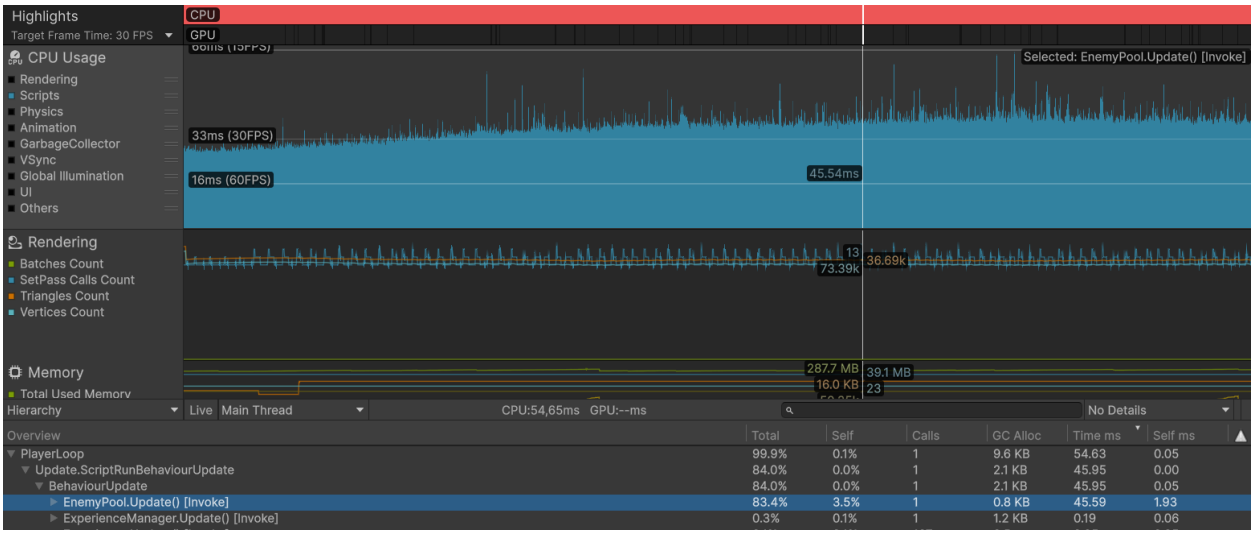
With 6000 enemies, the execution time now takes 19.64ms, the FPS in-game is around 60 which is an acceptable number. The player would probably still be able to play and not be too annoyed by the frame drop due to the high number of things displayed on the screen. This is a sort of limit where it's accepted by the player as he knows that a lot of instances reduce performance.

2.3.4 | 8000 Enemies



With 8000 enemies, it now takes 35.40ms, the game is barely playable with 30 FPS, and the player would probably still accept that if it is **temporary.** We are now close to the limit and we can predict that the next test will be the last as the game might become unplayable.

2.3.5 | 10000 Enemies



To update 10000 Enemies, it now takes 45.59ms, the in-game FPS is now below 20 FPS which makes it unplayable as expected. We can say that 10000 enemies is **the limit for the OOP implementation.**

2.4 Performance Summary table

	2000 Enemies	4000 Enemies	6000 Enemies	8000 Enemies	10000 Enemies
Execution time (ms)	4,21	8,94	19,64	35,4	45,59
Scaling Factor	0	2,12	2,20	1,80	1,29
Time per Enemy (ms)	0,0021	0,0022	0,0033	0,0044	0,0046

In a Survivor-like game, the player's tolerance for performance drops is implicitly linked to the visual intensity on screen. At a scale of 10,000 active enemies, the system maintains an execution time of 45.59 ms. While this exceeds the ideal 16.6 ms window for 60 FPS, this drop to approximately 20 FPS marks the functional limit of the OOP implementation.

The Time per Enemy remains highly efficient, only increasing from 0.0021 ms to 0.0046 ms as the horde grows. This linear scalability, maintained by a custom Boids-inspired steering behavior and a spatial grid system, ensures that the performance remains predictable.

2.5 Memory Management & Garbage Collection

Optimization is not solely about CPU time; it also requires strict memory monitoring. Implementing the Object Pool pattern drastically reduces dynamic allocations within the heaviest game loops. As demonstrated by the profiler, our critical path for enemy updates generates an extremely low allocation

of roughly **0 - 1KB** per frame. At this scale, this micro-allocation is negligible. Its impact on Garbage Collector triggers is practically non-existent, ensuring a smooth experience free from stuttering or lag spikes. Isolating and measuring this behavior validates the stability of the OOP architecture. Furthermore, it proves that the performance bottleneck encountered at 10,000 entities is primarily due to memory fragmentation rather than excessive Garbage Collection overhead.

2.6 Conclusion

This first phase successfully pushed standard Unity tools and classic OOP architecture to their absolute limits. By applying strict engineering principles, such as a spatial grid, object pooling, optimized Boids algorithms, and clean memory management, the game maintains a playable framerate with a massive horde of 8,000 active entities. However, a clear technical ceiling is reached at 10,000 entities, where the framerate drops to approximately 20 FPS.

This is, I believe, a robust and optimized result for an OOP project. Yet, to meet high-level production standards, where low-level optimization and massive scale are critical, this approach reveals its fundamental hardware limitations. The underlying issue is not the algorithmic logic itself, but how OOP inherently scatters data across the RAM. This memory fragmentation leads to constant Cache Misses, starving the CPU and preventing it from computing at its full potential.

To break this performance ceiling, drop below the 16.6 ms frame time at a massive scale, and truly maximize hardware efficiency, a paradigm shift is required. We must abandon the object-oriented logic in favor of a Data-Oriented approach that aligns with how the CPU cache operates. This is the exact purpose of the upcoming DOTS/ECS implementation.

3. DOTS/ECS

3.1 DOTS/ECS Phase Overview

After reaching the limits of the OOP implementation at around 10,000 enemies, the second phase of this project re-implements the enemy simulation using Unity's Data-Oriented Technology Stack (DOTS): Entities (ECS), the C# Job System and the Burst Compiler.

In ECS, an enemy is no longer a GameObject with MonoBehaviours, but an Entity: a simple identifier linked to a set of components that only contain data (position, speed, damage...). Entities sharing the same set of components are stored contiguously in 16 KB memory blocks called archetype chunks. The logic lives in Systems that read these arrays linearly, which is exactly how the CPU cache prefers to access memory.

The Job System then splits this work across all CPU cores, and Burst compiles the jobs into highly optimized native code. Together, these three tools target the exact root cause identified in the OOP phase: data scattered in memory and a simulation running on a single thread.

I chose a hybrid approach: only the massive part of the game (enemies and experience orbs) was moved to ECS, while the player, the weapons and the UI remain classic MonoBehaviours. Converting objects that only exist once would bring no measurable benefit, and would make the gameplay code harder to iterate on.

3.2 DOTS/ECS Implementation Choices

Spatial hash grid (EnemyHashSystem): The spatial grid from the OOP phase was rebuilt as a NativeParallelMultiHashMap, filled every frame by a parallel Burst job. Its capacity grows with the horde and shrinks back when the number of enemies drops sharply, which avoids a native reallocation every frame.

Parallel horde movement (EnemyMovementSystem): The same Seek + Separation steering behavior as in the OOP version, now executed as a Burst-compiled IJobEntity on all worker threads. The optimizations from the OOP phase were kept: the separation is time-sliced over 4 frames, limited to 4 neighbors, and the neighboring cells are visited in a spiral order to avoid any directional bias in the horde.

Entity Command Buffers: Creating or destroying entities (structural changes) is not allowed inside parallel jobs. Enemy deaths and experience orb spawns are therefore recorded in Entity Command Buffers and played back once, at the end of the simulation.

GPU-driven rendering: Enemies are rendered by Entities Graphics, which relies on the BatchRendererGroup API to draw hundreds of thousands of instances in a few dozen batches.

EnemyBridge facade: Weapons and projectiles remain MonoBehaviours and never use the ECS API directly. They go through a static facade, EnemyBridge, exposing simple methods such as TryGetClosestEnemy() and DealDamage(). The spatial grid and the command buffers stay hidden from the gameplay code.

Safe enemy destruction: Since command buffers are only played back at the end of the frame, an enemy hit by two projectiles during the same frame could be destroyed twice, dropping two experience orbs and throwing an error during playback. Each enemy now carries an enableable EnemyDeadTag component, switched on as soon as its destruction is queued, so that every code path (weapons, contact with the player, spawner) skips enemies that are already dying.

In-game benchmark controller: A small HUD and keyboard shortcuts allow changing the target horde size at runtime, from 1,000 up to 1,000,000 enemies, without recompiling or restarting the game.

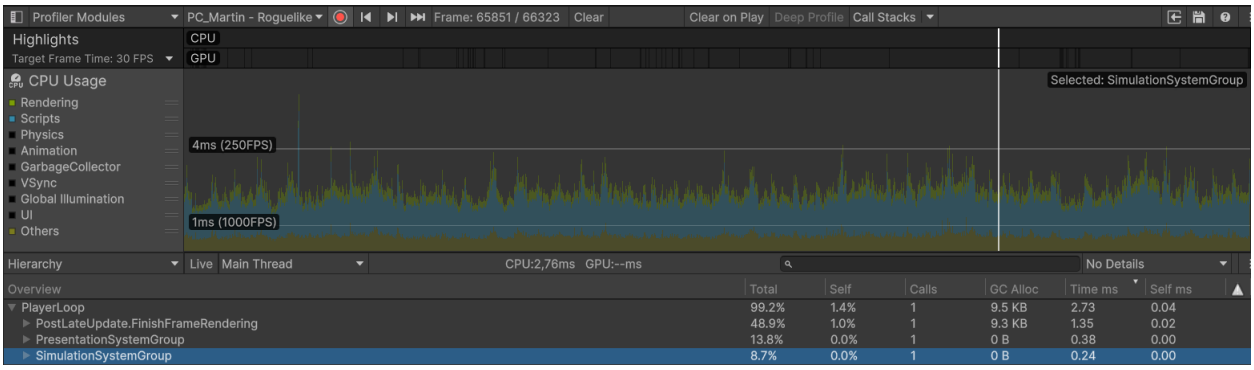
3.3 Performance analysis

To compare both implementations fairly, I measured the equivalent of EnemyPool.Update() from the OOP phase. In ECS, the enemy logic is split across several systems that all run inside the SimulationSystemGroup, so this group is the metric used below.

There is one important difference: the heavy work no longer runs on the main thread but on the worker threads. The time shown for SimulationSystemGroup on the main thread therefore includes scheduling the jobs and waiting for them to complete. It also includes the experience orbs and the spawner. This makes it a slightly pessimistic, but honest, equivalent of the OOP measurement.

I also report the PlayerLoop time, which represents the whole CPU frame of the game (simulation, rendering preparation, UI...). Both implementations were measured on the same machine, in a development build connected to the Unity Profiler, and each value comes from a representative frame once the horde size was stable.

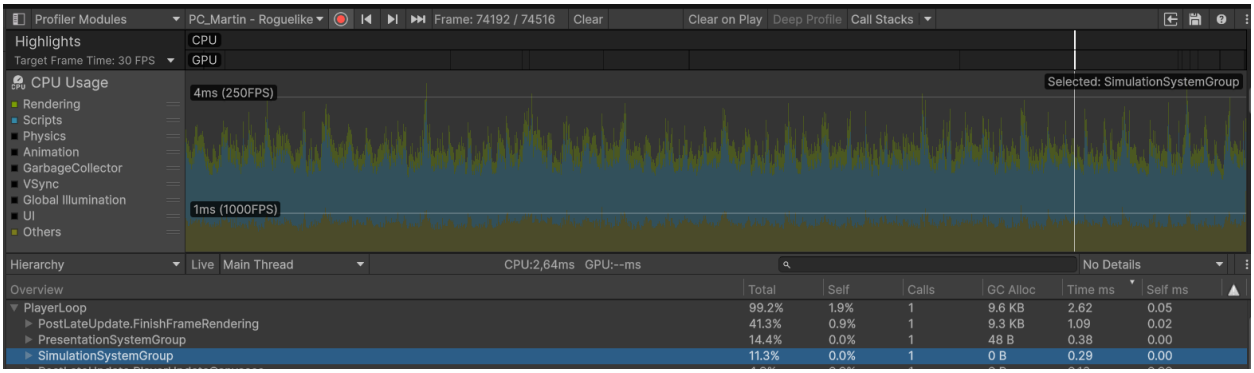
3.3.1 | 2,000 Enemies



With 2,000 enemies, the simulation takes 0.24 ms, compared to 4.21 ms with the OOP implementation.

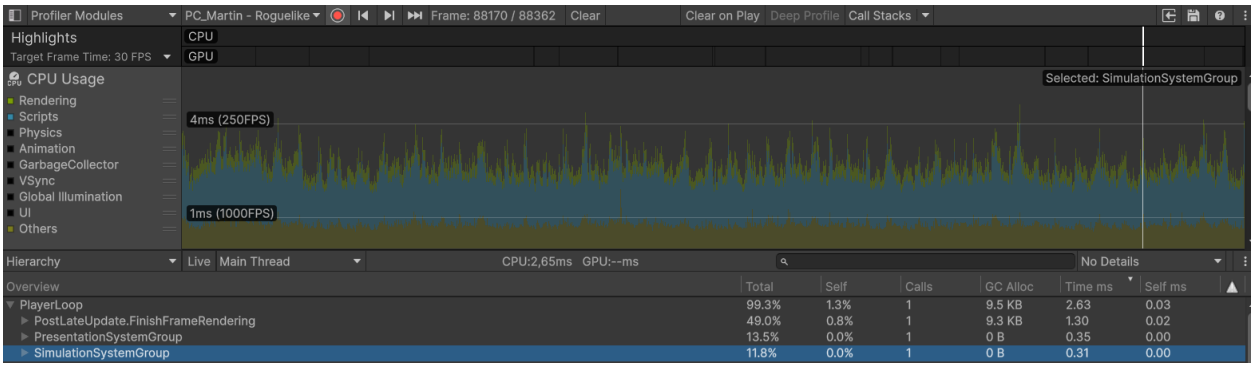
The whole frame only takes 2.73 ms on the CPU.

3.3.2 | 4,000 Enemies



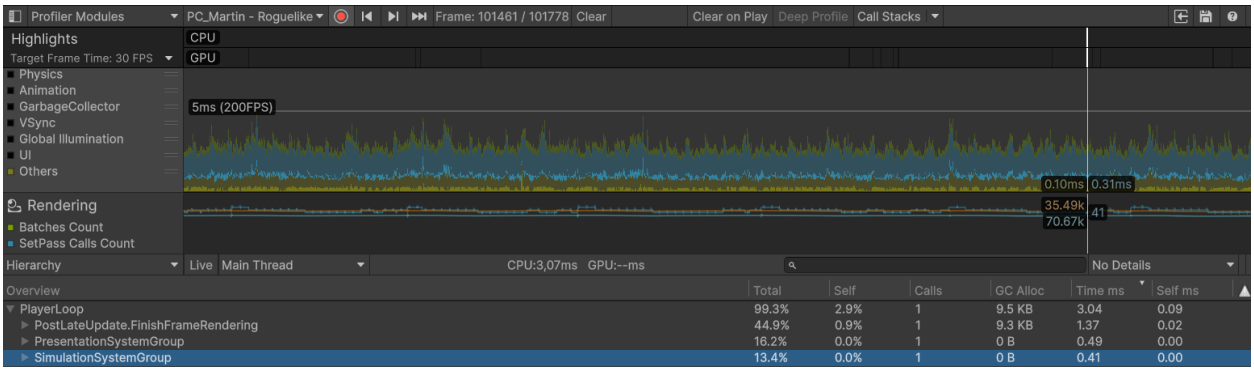
With 4,000 enemies, the simulation takes 0.29 ms (8.94 ms in OOP). Doubling the horde barely changes anything: at this scale, most of the cost is a fixed overhead (updating the systems, scheduling the jobs, synchronizations) rather than the enemies themselves.

3.3.3 | 6,000 Enemies



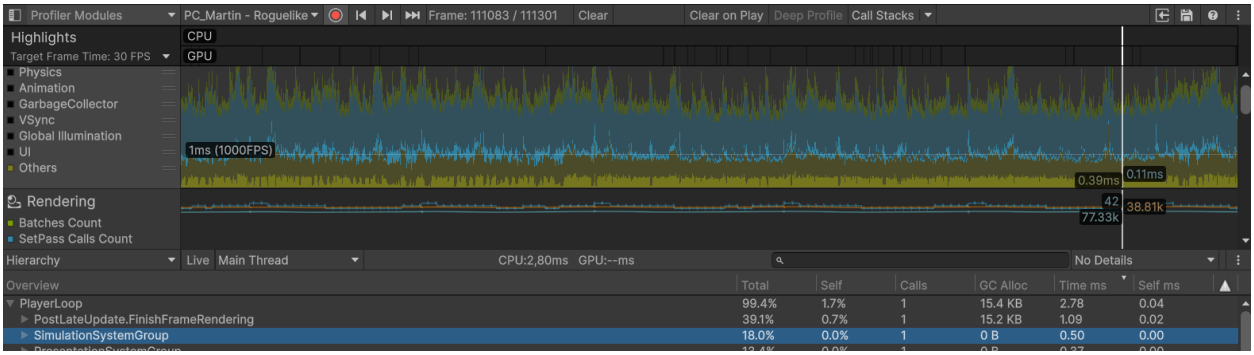
With 6,000 enemies, the simulation takes 0.31 ms (19.64 ms in OOP), while the OOP version was already down to around 60 FPS at this point.

3.3.4 | 8,000 Enemies



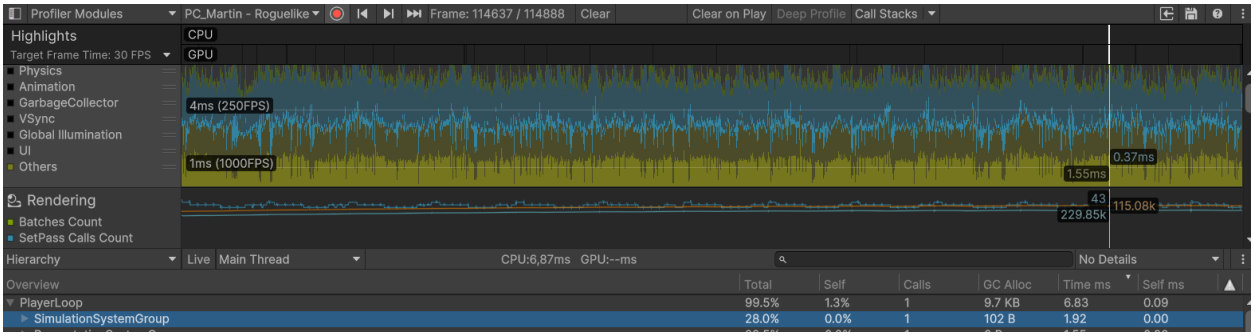
With 8,000 enemies, the simulation takes 0.41 ms (35.40 ms in OOP). Where the OOP implementation was barely playable at 30 FPS, the whole DOTS frame still only takes 3.04 ms on the CPU.

3.3.5 | 10,000 Enemies



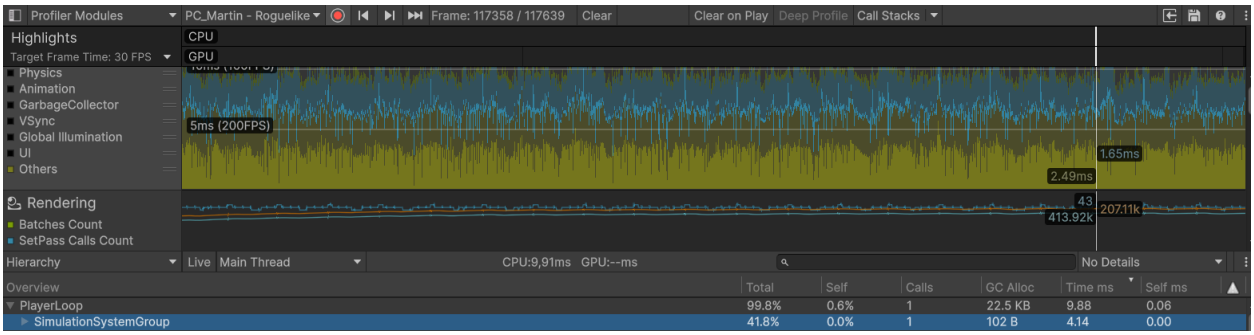
This is where the OOP implementation reached its limit: 45.59 ms for EnemyPool.Update() and below 20 FPS. With DOTS, the simulation of the same 10,000 enemies takes 0.50 ms, about 91 times faster, and the whole frame takes 2.78 ms. From this point on, there is no OOP equivalent anymore, so let's see how far the new implementation can go.

3.3.6 | 50,000 Enemies



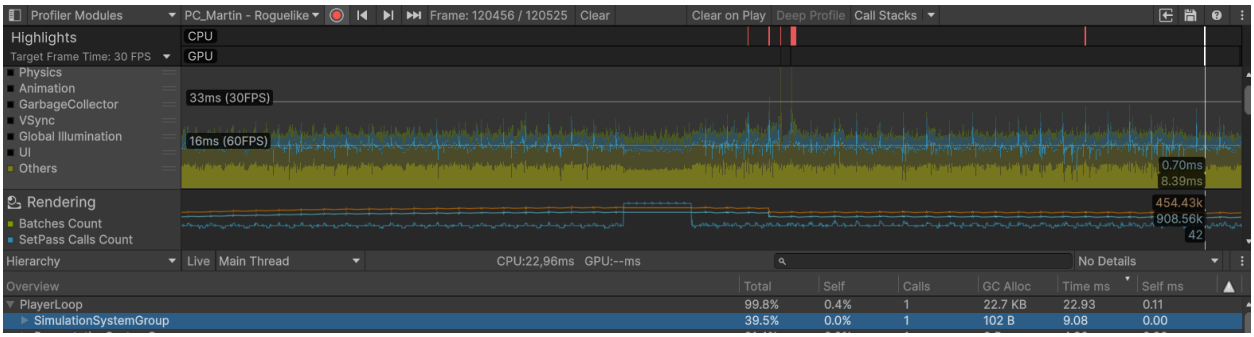
With 50,000 enemies, the simulation takes 1.92 ms and the whole frame 6.83 ms. The cost of each enemy now becomes visible: from here, the simulation time grows linearly with the horde size, at around 0.04 µs per enemy.

3.3.7 | 100,000 Enemies



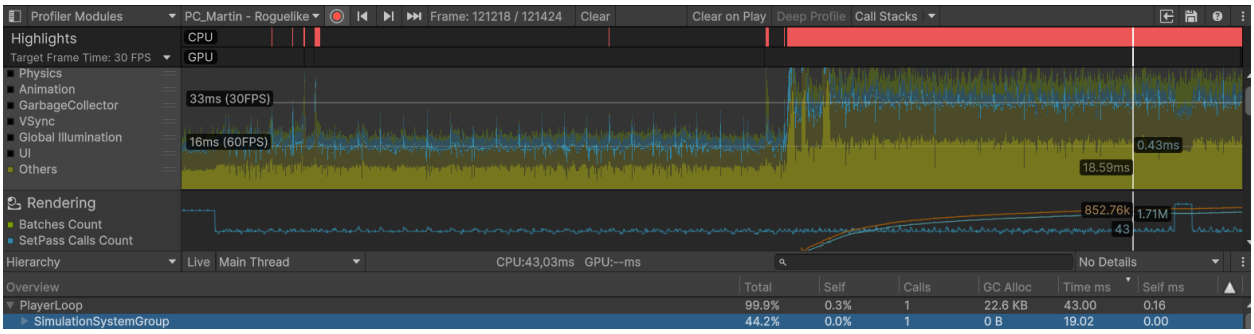
With 100,000 enemies, ten times the OOP limit, the simulation takes 4.14 ms and the whole frame 9.88 ms. The CPU can still deliver around 100 FPS.

3.3.8 | 250,000 Enemies



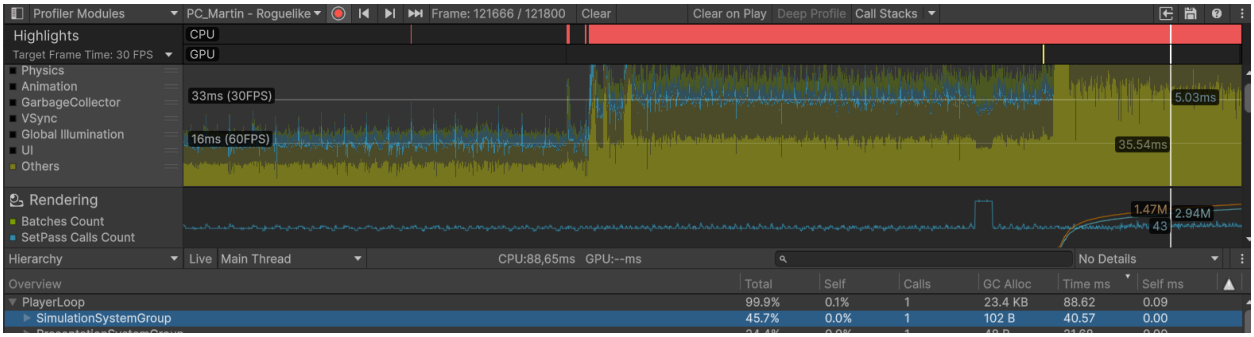
With 250,000 enemies, the simulation takes 9.08 ms and the whole frame 22.93 ms, around 43 FPS. This is the last tested step that stays above the 30 FPS threshold, which I consider the practical limit of the DOTS implementation.

3.3.9 | 500,000 Enemies



With 500,000 enemies, the simulation takes 19.02 ms and the whole frame 43.00 ms, around 23 FPS: the game drops below 30 FPS. The spikes visible at the beginning of the capture correspond to the spawning of the new enemies, which runs on the main thread in batches of up to 25,000 enemies per frame.

3.3.10 | 1,000,000 Enemies



With one million enemies, the simulation takes 40.57 ms and the whole frame 88.62 ms, around 11 FPS. The game is no longer playable, but the simulation of one million enemies still takes less time than the OOP implementation needed for 10,000 of them (45.59 ms).

3.4 Performance Summary table

Enemies	Simulation (ms)	Time per Enemy (µs)	Full CPU frame (ms)	CPU-bound FPS
2,000	0.24	0.120	2.73	~366
4,000	0.29	0.073	2.62	~382
6,000	0.31	0.052	2.63	~380
8,000	0.41	0.051	3.04	~329
10,000	0.50	0.050	2.78	~360
50,000	1.92	0.038	6.83	~146
100,000	4.14	0.041	9.88	~101

Enemies	Simulation (ms)	Time per Enemy (µs)	Full CPU frame (ms)	CPU-bound FPS
250,000	9.08	0.036	22.93	~44
500,000	19.02	0.038	43.00	~23
1,000,000	40.57	0.041	88.62	~11

Up to 10,000 enemies, the fixed overhead of the ECS pipeline dominates, which is why the first steps look almost flat. From 50,000 enemies onwards, the time per enemy stabilizes between 0.036 and 0.041 µs: the implementation scales linearly.

The FPS column is an estimate based on the CPU frame time only (1000 / PlayerLoop time). The GPU time was not captured, so the real framerate can be slightly lower.

3.5 OOP vs DOTS/ECS Comparison

Enemies	OOP EnemyPool.Update() (ms)	DOTS SimulationSystemGroup (ms)	Speed-up
2,000	4.21	0.24	×17.5
4,000	8.94	0.29	×30.8
6,000	19.64	0.31	×63.4
8,000	35.40	0.41	×86.3
10,000	45.59	0.50	×91.2

Two numbers summarize this comparison. Per enemy, the simulation cost drops from about 4.6 µs (OOP, 10,000 enemies) to about 0.04 µs (DOTS, 1,000,000 enemies), roughly 110 times less. On the whole CPU frame with 10,000 enemies, the game goes from 54.63 ms to 2.78 ms, about 20 times faster. The second number is smaller because the rest of the frame (rendering, UI, engine) did not benefit from the migration.

In practice, the playable limit moves from about 8,000 enemies at 30 FPS to about 250,000 enemies at more than 40 FPS: around 30 times more enemies on screen.

These gains combine three effects that are not measured separately here: the contiguous memory layout, the Burst compilation and the multithreading. The rendering is also different (SpriteRenderer in OOP, Entities Graphics in DOTS), which contributes to the gain on the whole frame.

3.6 Memory Management & Garbage Collection

The SimulationSystemGroup allocates 0 B per frame up to 10,000 enemies, and only 102 B above 50,000 enemies. All the ECS data lives in unmanaged memory (components and native containers), which the Garbage Collector never has to scan.

The allocations visible on the PlayerLoop (around 9.5 KB per frame with 2,000 enemies) almost entirely come from PostLateUpdate.FinishFrameRendering (9.3 KB), in other words from the render pipeline and not from the game simulation. As in the OOP phase, garbage collection is not a factor in these results.

3.7 Conclusion

The DOTS/ECS implementation broke the ceiling reached in the OOP phase. By storing the enemies contiguously in memory, compiling the logic with Burst and spreading it across all CPU cores, the simulation of 10,000 enemies went from 45.59 ms to 0.50 ms, and the game now stays above 40 FPS with 250,000 enemies.

This phase also moved the bottleneck. With a large horde, the simulation only represents around 40 to 45% of the CPU frame; the rest is spent on rendering and engine overhead. Pushing the limits further would now mean optimizing the rendering side and moving the spawning logic into a Burst job, rather than optimizing the enemy logic itself.

Beyond the raw numbers, this project showed that DOTS is not a free performance switch: it requires designing around data, handling job dependencies and structural changes, and building bridges with the classic MonoBehaviour world. Used where it matters, on the thousands of identical entities of a Survivor-like, it turns a hard technical limit into a comfortable margin.